Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and Communications
Technology Department

Academic Year	2017-2018
Semester	First
Exam Date	23 /12 / 2017

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject:	(E501) Digital Signal Processing	**Spec.:**	5th Elec. Eng. & Comm. Tech. Dep.
Examiners:	Dr. Samir Kamal and Prof. Mostafa Afifi	**Time:**	3 hours

Number of Pages: 2 | Number of Questions: 5 | Attempt all questions

Question 1 (20 points)

a - In detail, what are the main characteristics of the discrete time sinusoidal signals?

b- Check the periodicity of the following sequences and find their fundamental periods.
(i) X (n) = 2 Sin (3 π n / 105). (ii) x (t) = cos (3000 π t).

(c) For the signal $x(t) = 2\cos(2000\pi t)$ to be sampled with sampling frequency F_S = 2 KHz.

(i) Is F_S suitable for sampling the given signal, $x(t)$,
(ii) If it is suitable, use it to find an expression for the corresponding sequence, x(n),
(iii) Find the values of x(n) in the first cycle and
(iv) Plot (showing the steps) the given $x(t)$ and the obtained x(n) on the same figure.

Question 2 (20 points)

a- Find the relationship between the frequency of analog signal, frequency of discrete-time signal and sampling frequency.

b- Show that the DFT, X(K), of a sequence x(n) is periodic.

c- For the following sequence: x(n) = {1, 0, 0, 1}:

(i) Using the direct method, DFT, find the frequency components of this sequence.
(ii) Check your results obtained in part (i) by calculating x(n) using the IFFT algorithm.
(iii) Using your results, verify Parseval's theorem.
(iv) Plot the amplitude and the phase spectrums.
(v) What is the computational saving if we use FFT algorithm instead of DFT algorithm.

Question 3 (15 points)

a- Define (using equations and/or figures) the following items: Correlation operation, Correlation coefficient and Convolution operation.

b - Given two sets of samples $x_1(n)$ = {3, 2, 0, 1} and $x_2(n)$ = {1, 2, 0, 1}.
(i) Find the linear cross- correlation between $x_1(n)$ and $x_2(n)$.
(ii) Plot the correlation function obtained in part (i) v.s. the lag number, j.
(iii) Find the autocorrelation function of the sequence $x_2(n)$.

c- For the following sequences, x1 (n) = {3, 2, 0, 1} & x2 (n) = {1, 2, 0, 1}, Find the convolution of them and plot this convolution.

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Question 4 (20 points)

a- For LTIS define (using equations and/or figures) the linearity, stability, ROC and difference equation.

b- For the following transfer function, *H(Z),* use the power series method find its impulse response, h(n).

$$H(z) = \frac{z^2(z+1)}{(z-0.5)(z-1)^2}$$

c- For the transfer function of a digital system given in part (b):

(i) Plot the pole-zero diagram of H(z),
(ii) Discuss the system stability,
(iii) Determine the ROC of this transfer function,
(iv) Obtain the difference equation of this system,
(v) Represent the filtering operation in a block diagram form.

Question 5 (15 points)

a- Given the realization structure of an IIR filter as shown in Fig. 1.

(i) Find the two-step difference equation relating the output, y (n), and input, x (n).
(ii) Draw the equivalent Lattice filter's structure, and then
(iii) Compute its coefficients.

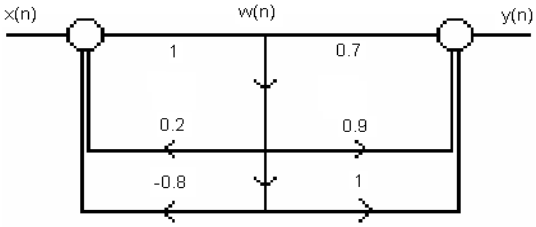


Fig. 1

b- Check (✔) for the right statements and (X) for the wrong statements and correct the mistakes.

i- In quantization process, when the quantization step (Δ) increases, the accuracy of the quantizer increases. ( )
ii- Nyquist frequency (F_N), is the minimum sampling frequency ( )
iii- Fourier transform (FT) is calculated for a non-periodic continuous – time signals. ( )
iv- Discrete Fourier transform (DFT) is obtained for the analog signals. ( )
v- To reconstruct the continuous time signal, x (t), from its discrete time sequence, X(n), the sampling rate, F_S, must be less than the frequency of x(t) signal, f_0. ( )

Good Luck